Campus Deployment Strategy

Campus Wastewater Health Alert Dashboard

Objective: Deploy a wastewater-based epidemiology (WBE) system across the campus to provide early detection of infectious disease outbreaks and sanitation incidents through routine wastewater monitoring.

1. Deployment Architecture

- Install sampling points at major sewer outlets serving Academic Blocks, Boys Hostels, Girls Hostels and Mess facilities.
- Use automatic or manual composite samplers to collect wastewater once every 24 hours.
- Transport samples to the campus laboratory for analysis.
- Upload laboratory results to the dashboard for automated risk assessment and visualization.

2. Sampling Strategy

- Sampling Frequency: Daily under normal operation.
- Yellow Alert: Increase sampling to twice per day at the affected location.
- Red Alert: Daily sampling at the affected site plus neighbouring buildings until the alert is cleared.
- Measure viral markers, bacterial indicators, physicochemical parameters and antibiotic residues.

3. Data Processing Pipeline

- Collect wastewater samples.
- Perform laboratory analysis.
- Upload results to the dashboard.
- Normalize viral markers using wastewater flow.
- Compare values against predefined thresholds.
- Generate Health Score, Alert Level, Recommendations and Risk Trend.
- Notify campus authorities.

4. Alert Response Strategy

- Green: Routine monitoring and preventive maintenance.
- Yellow: Increase sampling frequency, inspect sewer infrastructure and notify the campus health centre.
- Red: Confirm through re-sampling, inspect neighbouring buildings, initiate targeted health advisory, increase sanitation measures and coordinate with public health authorities when required.

5. Hardware Requirements

- Automatic wastewater samplers
- Flow meter
- pH and turbidity sensors
- Cold storage for sample transport
- Laboratory testing equipment (PCR/microbiology)
- Internet-connected workstation running the Streamlit dashboard

6. Software Infrastructure

- Python
- Streamlit Dashboard
- Pandas & NumPy
- Plotly Visualizations
- GitHub for version control
- Secure campus server or cloud deployment

7. Privacy and Ethics

- Monitor buildings instead of individuals.
- No personally identifiable information is collected.
- Alerts are initially shared only with authorized campus health officials.
- Public communication occurs only after verification.

8. Deployment Phases

- Phase 1 – Pilot: Deploy at selected hostels and one mess.
- Phase 2 – Campus Expansion: Cover all academic, hostel and mess locations.
- Phase 3 – Continuous Monitoring: Daily surveillance, forecasting and periodic calibration.

9. Expected Benefits

- Early outbreak detection before widespread clinical reporting.
- Faster response to sanitation failures.
- Data-driven campus health management.
- Reduced spread of infectious diseases.
- Improved public health preparedness.

10. Future Scope

- IoT-enabled real-time sensors
- Machine learning based outbreak prediction
- Integration with weather and sewer network models
- Mobile notification system
- Multi-campus deployment

Conclusion

The proposed deployment strategy enables a scalable, privacy-preserving wastewater surveillance system for continuous campus health monitoring. By combining systematic sampling, laboratory testing, automated analytics, and structured response protocols, the platform can provide actionable early warnings for infectious disease outbreaks and environmental contamination while supporting informed decision-making by campus authorities.